

Chunlin **Bian**

Target roles: Embedded Applications / Embodied AI

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"Self-discipline and social commitment"

Profile

Architecture Ph.D. candidate at Tongji University with experience in research, vision-based sensing applications, and full-stack development. Implement algorithm code for video-based earthquake analysis and contribute UI design/code optimization to an embedded smart lamp. Interested in intelligent systems connecting perception, decision-making, and physical feedback.

Technical Skills

- **Engineering:** Python; JavaScript/TypeScript; React; Node.js; REST APIs; Git; Windows/Linux.
- **AI and data:** Agent workflows, RAG, multi-model orchestration; video response analysis; reinforcement learning and evacuation simulation; GIS/InSAR.
- **Hardware-facing projects:** Smart-lamp UI/code contributions; collaboration on camera input, state-machine logic, and light/voice feedback.

Selected Projects and Research

Video-Based Earthquake Detection and Response

Algorithm code and implementation | Windows / Linux; conventional cameras *2026*

- Research video-based earthquake detection and response analysis using conventional camera feeds; responsible for algorithm code and implementation in Windows and Linux environments.
- Contributed to Earthquake Response Analysis Software Based on Surveillance Video V1.0; joint copyright holder, registered in September 2026 (2026SR0931084).

Randeng Genie | Embedded Smart Desk Lamp

INTERCITY HACK, Ningbo | Visual/UI design and code optimization *30 Oct–1 Nov 2026*

- Contributed visual/UI design and code optimization to a child-focused smart desk lamp combining posture reminders, facial-expression cues, and light/voice feedback.
- Aligned interface design and code improvements with the team's camera, five-state FSM, WS2812 lighting, and offline voice architecture; on-device model deployment and hardware integration remain validation tasks.
- Participated in the Ningbo edition of INTERCITY HACK, a linked Hangzhou–Ningbo–Wenzhou hackathon focused on AI for industry and the real economy.

Seismic Resilience and Multi-Agent Evacuation

Tongji University | Doctoral research *2021–Present*

- Investigate multi-agent reinforcement learning versus classical models for hospital evacuation under earthquakes; first-author paper in Buildings (2026).
- Doctoral research combines GEM seismic-hazard and GHSL population data, GIS, AHP–entropy weighting, and recovery curves to assess urban seismic resilience.
- Incorporate spatiotemporal population exposure and multi-agent evacuation simulations to study interactions among built environments, human behavior, and urban-renewal interventions.

Software Development Projects

StockMate AI

Independent full-stack developer | AI investment research — *2024–Present*

- Independently designed and developed an AI investment-research product for older adults, covering frontend interactions, backend services, financial-data APIs, multi-model orchestration, and demos.
- Built a multi-agent workflow spanning macroeconomics, industries, financial statements, technical analysis, sentiment, risk, and strategy; used RAG to retrieve filings/news and generate structured reports.
- Implemented accessible market monitoring, financial Q&A, and strategy simulation; advanced to the national second round through the 2026 Tongji University AI Hackathon Tour event.

Professional Experience

Xichang University | Former Senior Lecturer

GIS, parametric design, and research software — *Joined Feb 2016 (former)*

- Taught GIS, parametric design, urban planning, and landscape architecture; led an InSAR subsidence-monitoring project and contributed to deformation-monitoring research.
- Contributed to laboratory development, two radar corner-reflector utility-model patents, and four software copyright registrations for InSAR and geospatial analysis.

Yiren Architecture and Landscape Design Co., Ltd.

Founder — *Jan 2018–Present*

- Led rural-revitalization, forest-park, and heritage-area planning projects, coordinating requirements, design iterations, spatial analysis, and delivery.

Panzhihua University | Lecturer in Architecture

Architecture, urban design, and landscape architecture — *Aug 2010–Feb 2016*

- Taught courses, supervised graduation projects, and contributed to industry collaborations and local architectural research.

Education

- **Tongji University** — Ph.D. candidate, Architecture, Sep 2021–Present; seismic resilience, multi-agent simulation, and urban renewal.
- **Massey University** — Visiting Scholar, 2022; CSC-funded research in digital spatial analysis and landscape ecology.
- **Peking University** — Master's degree, Landscape Architecture, 2008–2010.
- **Sichuan Agricultural University** — Bachelor's degree, Landscape Architecture, 2003–2007.

Selected Results and Recognition

- **Hongmeng team lead; Third Prize, final round** — 2026 China (Guangxi)–ASEAN AI + Emergency Management Innovation Competition, Track 5 (Universities and ASEAN), Sep 2026; AI-driven urban seismic-resilience sensing and analysis platform.
- **Joint software copyright holder** — Surveillance-video earthquake response analysis (2026SR0931084); City Radar and Data Technology Platform V1.0 (2026SR0999998).
- **First-author paper** — Bian C, Guo Y, Meng G, et al. Modeling seismic resilience and hospital evacuation: A comparative analysis of multi-agent reinforcement learning and classical evacuation models. *Buildings*, 2026, 16(8): 1538.
- **First-author paper, accepted** — Bian C, Meng G, Hu B, Yang W. The Construction of "Republican-Era Style" in the Renewal of Yuantian Street, Huili Ancient City: Contemporary Translation of Modern Historical Imagery. *Housing Science* (in Chinese); accepted 29 Oct 2026, provisionally scheduled for issue 12, 2026.
- **First-author conference paper, planned publication** — Bian C, Meng G, Yang J. Structural Transformation and Functional Reconstruction in a Modern Historical Context: A Study of the Bell and Drum Tower in Huili Ancient City, Sichuan. Listed in the contents of Research and Conservation of Modern Chinese Architecture, Vol. 14; planned with Tianjin University Press (in Chinese).
- **Young Geographers Award — Best Paper**, 15th Korea–China–Japan Joint Conference on Geography, 2023.